

Chapter 3: Plate Tectonics

Physical and Structural Geology (GS 21/L)

Engr. Karl Patrick B. Garcia

Intended Learning Outcomes

- Discuss basic geological concepts and theories in understanding Earth's history, specifically the Continental Drift Theory and the Seafloor Spreading Concept.
- Analyze the principles of Plate Tectonics, recognizing the distribution of major lithospheric plates.
- Interpret the formation mechanisms and characteristics of the three main types of plate boundaries: Divergent, Convergent, and Transform.
- Evaluate the primary mechanisms driving plate motion, including slab pull, ridge push, and mantle convection models.
- Identify and analyze the Philippine Tectonic Setting, distinguishing the major tectono-stratigraphic blocks and regional fault zones.

Lithosphere & Asthenosphere

The foundation of plate tectonics relies on the mechanical distinction between Earth's outer layers.

- The Lithosphere: The rigid outer part of the Earth, consisting of the crust and upper mantle. This layer is broken into numerous distinct, rigid plates.
- The Asthenosphere: The upper layer of the Earth's mantle, situated directly below the lithosphere.
- In the asthenosphere, there is relatively low resistance to plastic flow, allowing it to behave like a highly viscous, ductile solid.
- This physical property differential allows the rigid lithospheric plates to move independently over the underlying asthenosphere.

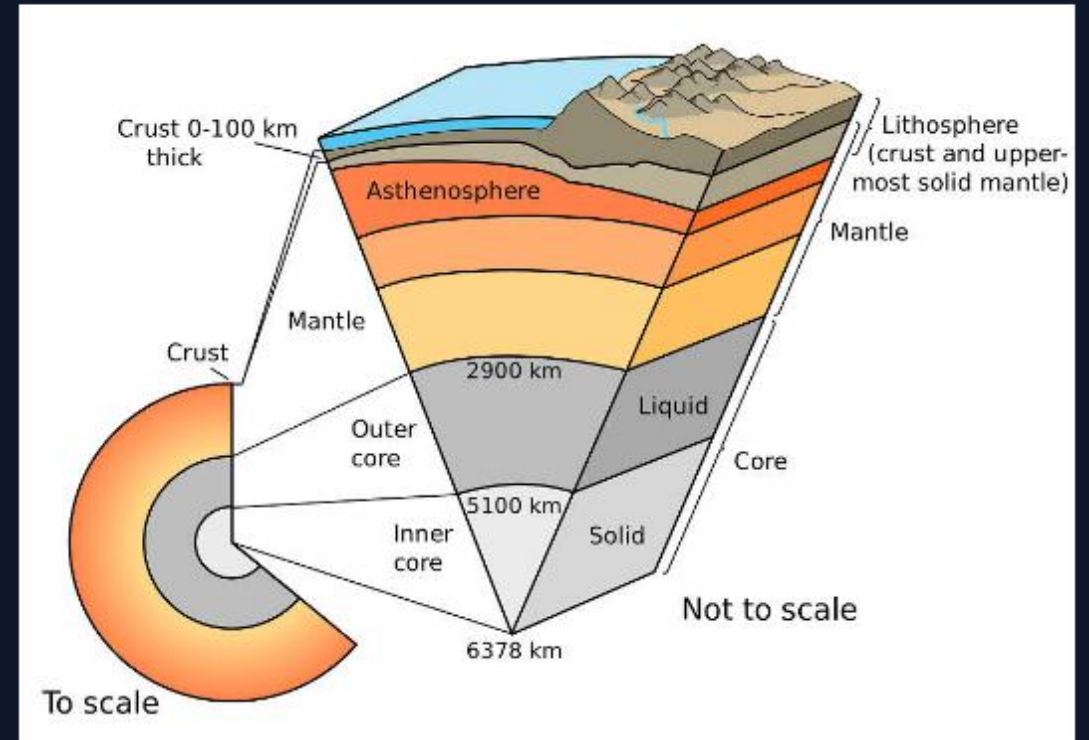
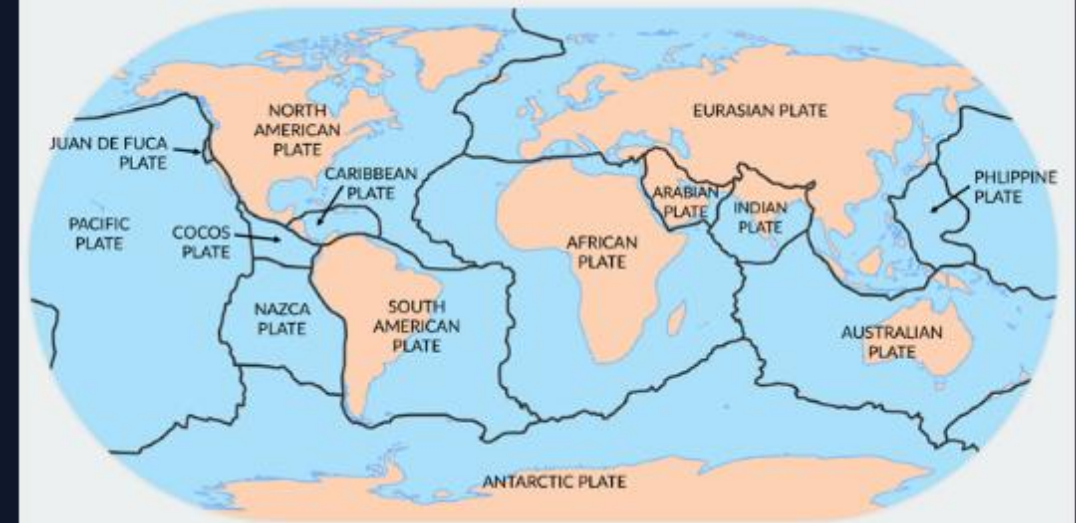


Plate Tectonics Theory

The Plate Tectonics Theory acts as the unifying model of Earth sciences, proposing that the lithosphere is divided into numerous tectonic plates that are in continuous, slow motion.

There are seven major lithospheric plates recognized globally:

- North American Plate
- South American Plate
- Pacific Plate (The largest, almost entirely oceanic)
- African Plate
- Eurasian Plate
- Australian-Indian Plate
- Antarctic Plate



Summary of Plate Boundaries

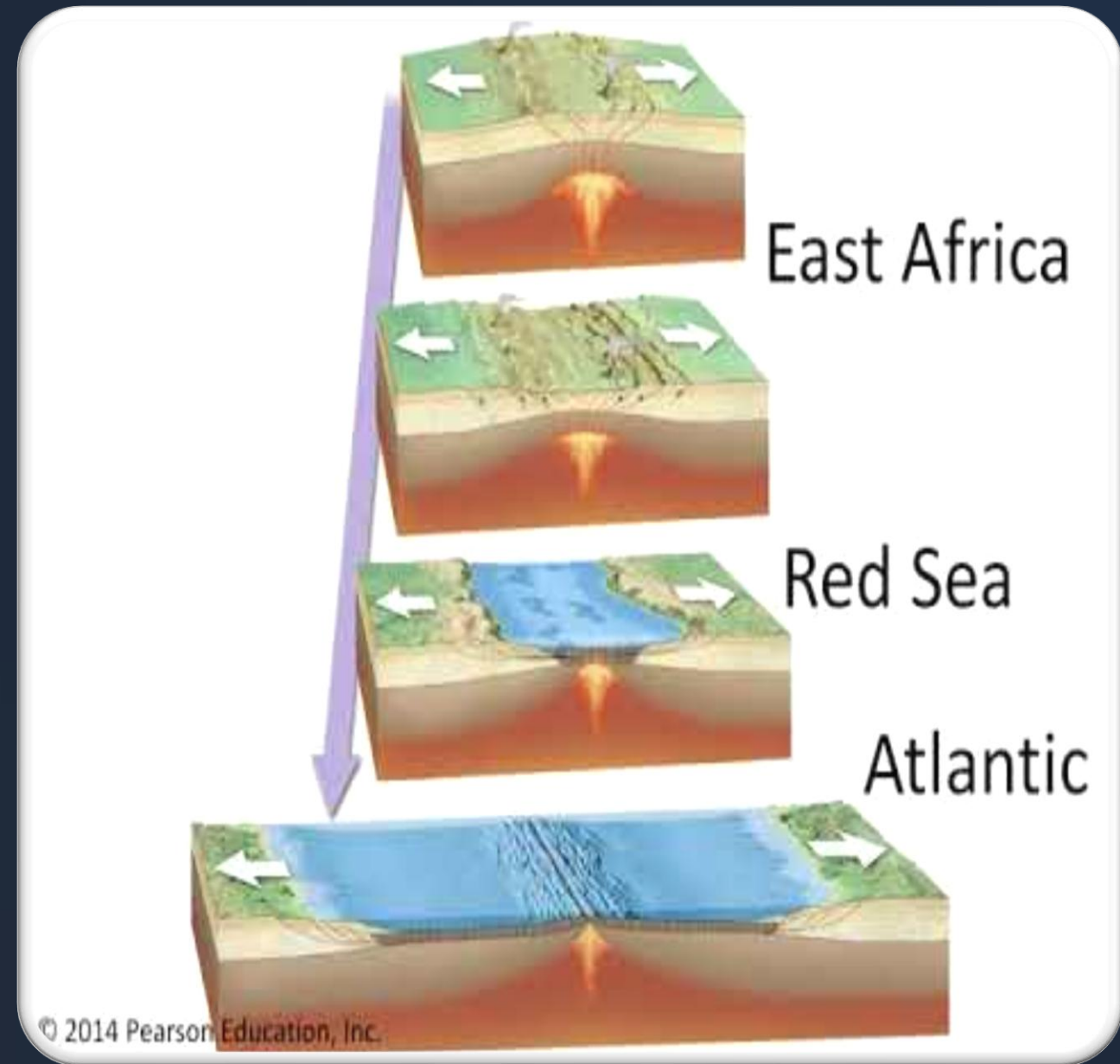
Boundary Type	Plate Motion	Topographic Effect	Volcanic Activity
Divergent	Spreading (moving apart)	Constructive (oceanic ridge/rift valley)	Yes
Convergent	Subduction (moving together)	Destructive (oceanic trench)	Yes
Transform	Lateral Sliding (past each other)	Conservative (no major relief)	No

Note: Plate boundaries are not static; they migrate and change over geologic time as plates are continually created and destroyed.

Divergent Boundaries

Divergent plate boundaries are constructive margins where two adjacent plates pull apart from each other.

- **Upwelling Magma:** As plates separate, fractures are created. Hot mantle rock ascends to fill the void, generating magma through decompression melting.
- **New Crust:** This magma cools and crystallizes to form new oceanic lithosphere.
- **Topography:** Characterized by oceanic ridges (e.g., Mid-Atlantic Ridge) and continental rift valleys.
- **Spreading Rates:** Typically range from 2 to 15 centimeters per year globally.



Evolution of a Divergent Boundary

Continental rifting occurs when divergent forces act on a continental landmass, ultimately leading to the formation of a new ocean basin.

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1 Continental Rift

Tensional forces stretch the crust, causing it to thin and fracture.

Large blocks drop down, creating a continental rift valley. *Example: East African Rift System.*

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2 Linear Sea

As spreading continues, the rift deepens and connects to the ocean, allowing seawater to flood the valley, forming a narrow linear sea. *Example: The Red Sea.*

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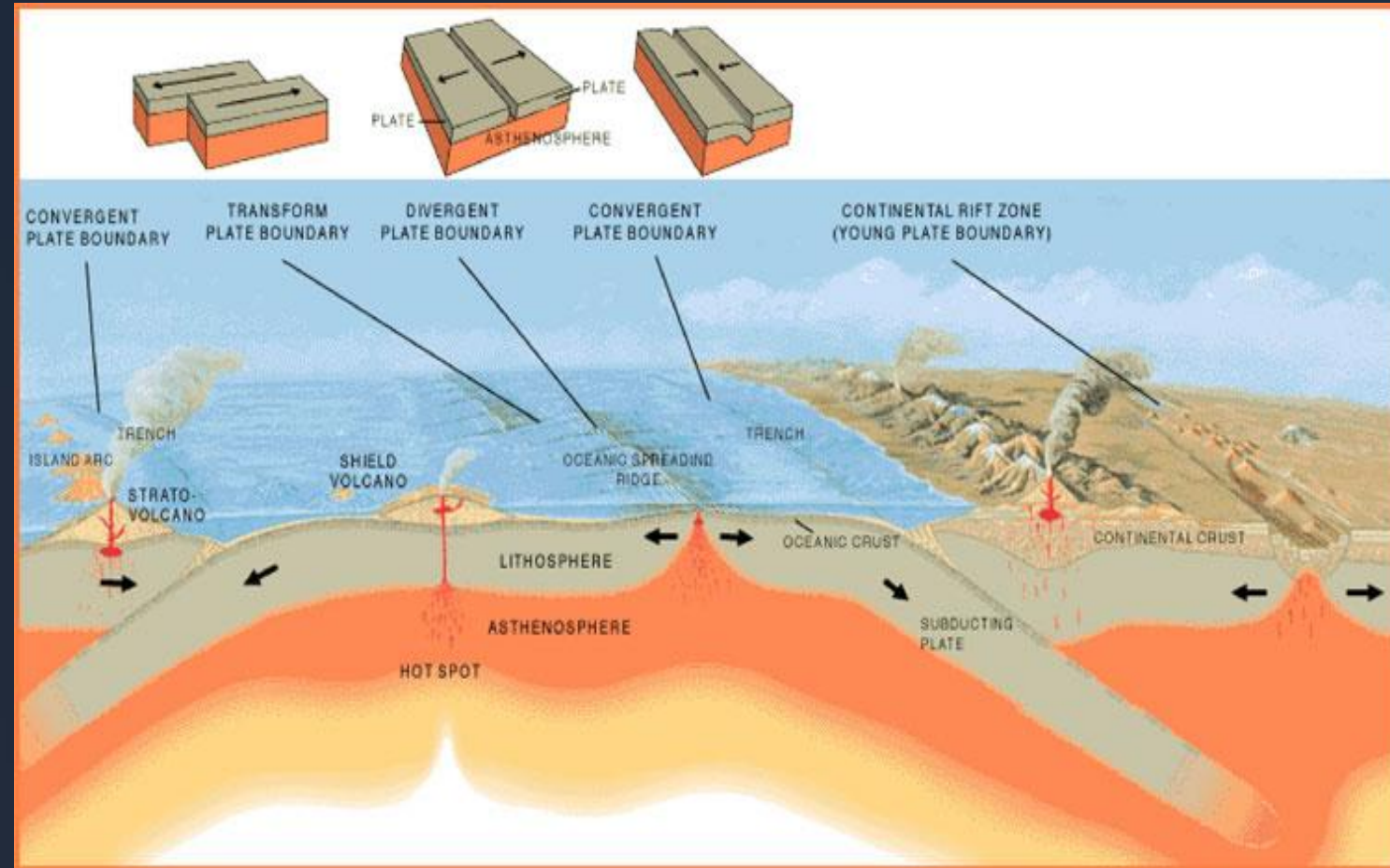
3 Mature Ocean Basin

Extensive spreading over millions of years produces a wide ocean basin featuring a central mid-ocean ridge and passive continental margins. *Example: Atlantic Ocean.*

Convergent Boundaries

Convergent boundaries are destructive margins where two plates collide, resulting in the consumption of lithosphere.

- Subduction Zones: Older, denser oceanic lithosphere plunges beneath an overriding plate and descends into the mantle.
- Deep-Ocean Trenches: Surface manifestations of subduction, forming the deepest parts of the ocean floor.
- Volcanism: Subduction introduces water to the hot mantle, triggering partial melting and generating magma that rises to form volcanic arcs.



Types of Convergent Boundaries

Oceanic-Continental

Denser oceanic slab subducts beneath continental crust. Magma ascends to form a continental volcanic arc. *Example: The Andes Mountains.*

Oceanic-Oceanic

One oceanic slab subducts beneath another. Magma generation produces a chain of volcanic islands known as a volcanic island arc. *Example: The Philippine Archipelago.*

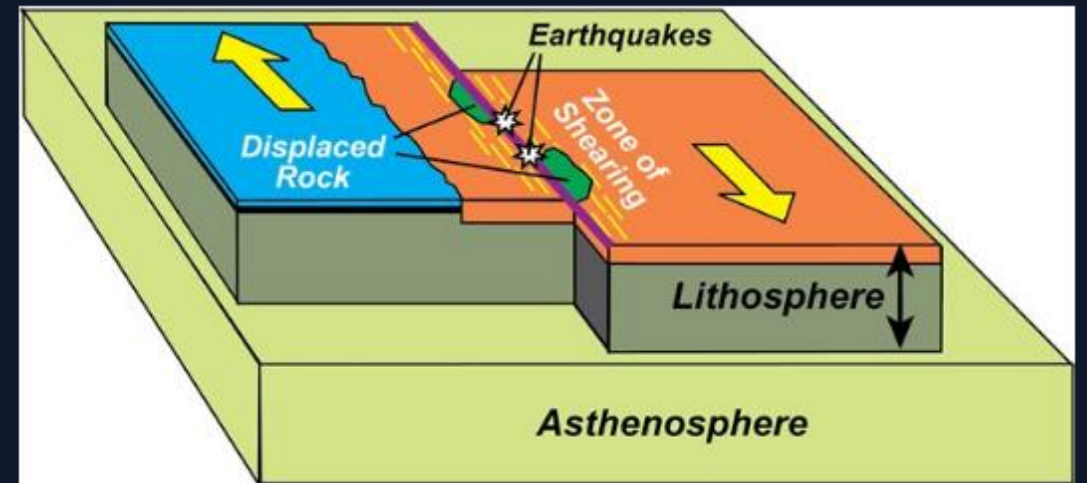
Continental-Continental

Continued subduction brings two continents together. Buoyant continental crust resists deep subduction, leading to a massive collision and mountain building. *Example: The Himalayas.*

Transform Fault Boundaries

Transform boundaries are conservative margins where plates slide horizontally past one another.

- **No Lithosphere Altered:** At transform faults, lithosphere is neither created nor destroyed.
- **Fracture Zones:** Most transform faults are located on the ocean floor, where they offset segments of the oceanic ridge system.
- **Continental Transforms:** A few large transform faults cut through continental crust, accommodating plate motion and causing severe shallow earthquakes.
- *Example: The San Andreas Fault in California and the Philippine Fault Zone.*

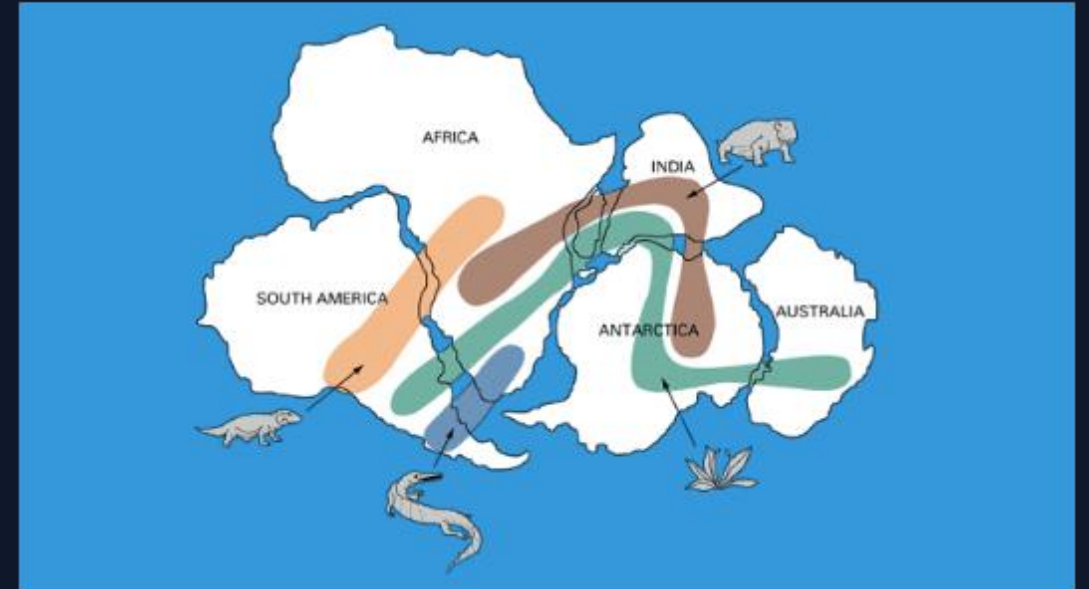


Continental Drift Theory

Proposed by Alfred Wegener in 1915, this hypothesis argued that the continents were once united in a supercontinent called Pangaea.

Wegener cited compelling observational evidence:

- The Continental Jigsaw: Remarkable morphological fit of coastlines, particularly South America and Africa.
- Fossil Evidence: Identical fossil organisms (e.g., *Mesosaurus*) discovered on landmasses separated by vast oceans.
- Rock Types & Structures: Matching geologic features, mountain belts, and rock ages across the Atlantic.
- Ancient Climates: Paleoclimatic data, such as glacial deposits found in modern tropical latitudes.



The Seafloor Spreading Concept

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New oceanic crust is continuously formed at mid-ocean ridges and spreads laterally, pushing older crust away like a conveyor belt.

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Proposed by Harry Hess in the early 1960s, this theory provided the crucial mechanism for plate motion that Wegener's Continental Drift hypothesis lacked, leading directly to the modern theory of Plate Tectonics.

Evidence: Ocean Drilling & Rock Ages

The Deep Sea Drilling Project

From 1968 to 1983, the Glomar Challenger collected hundreds of core samples from the ocean floor globally. This scientific expedition aimed to gather data to test the seafloor spreading hypothesis.

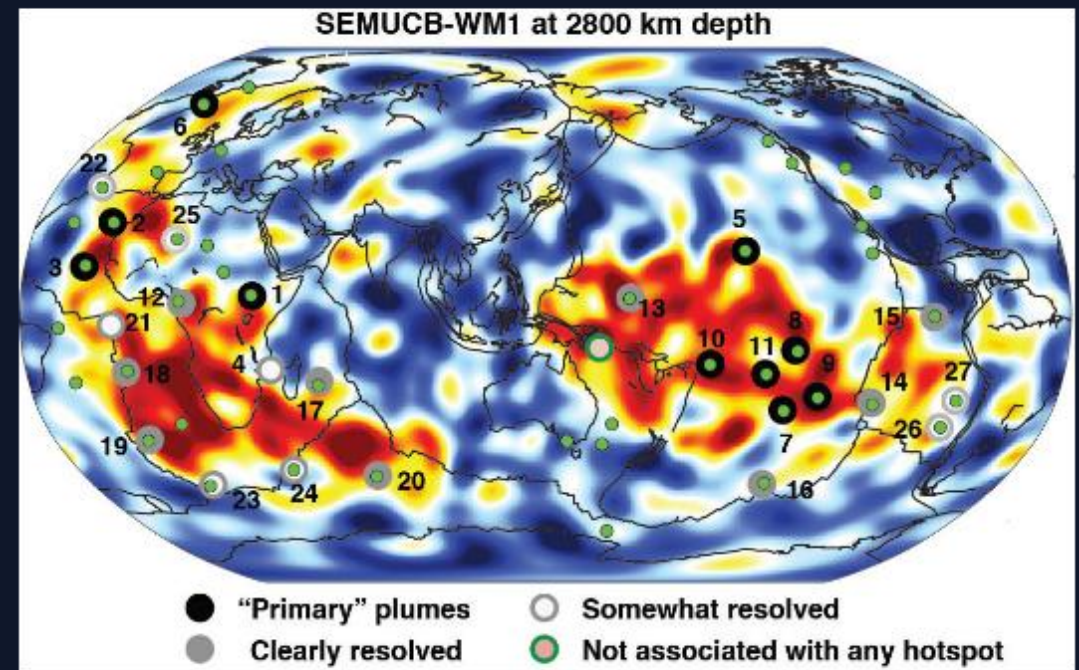
Age of the Seafloor

Sediment dating confirmed that the age of the oceanic crust increases proportionally with distance from the mid-ocean ridge. The oldest oceanic crust is roughly 180 million years old, compared to continental rocks exceeding 4 billion years.

Evidence: Mantle Plumes & Hotspots

A mantle plume is a cylindrically shaped upwelling of exceptionally hot rock originating deep within the mantle.

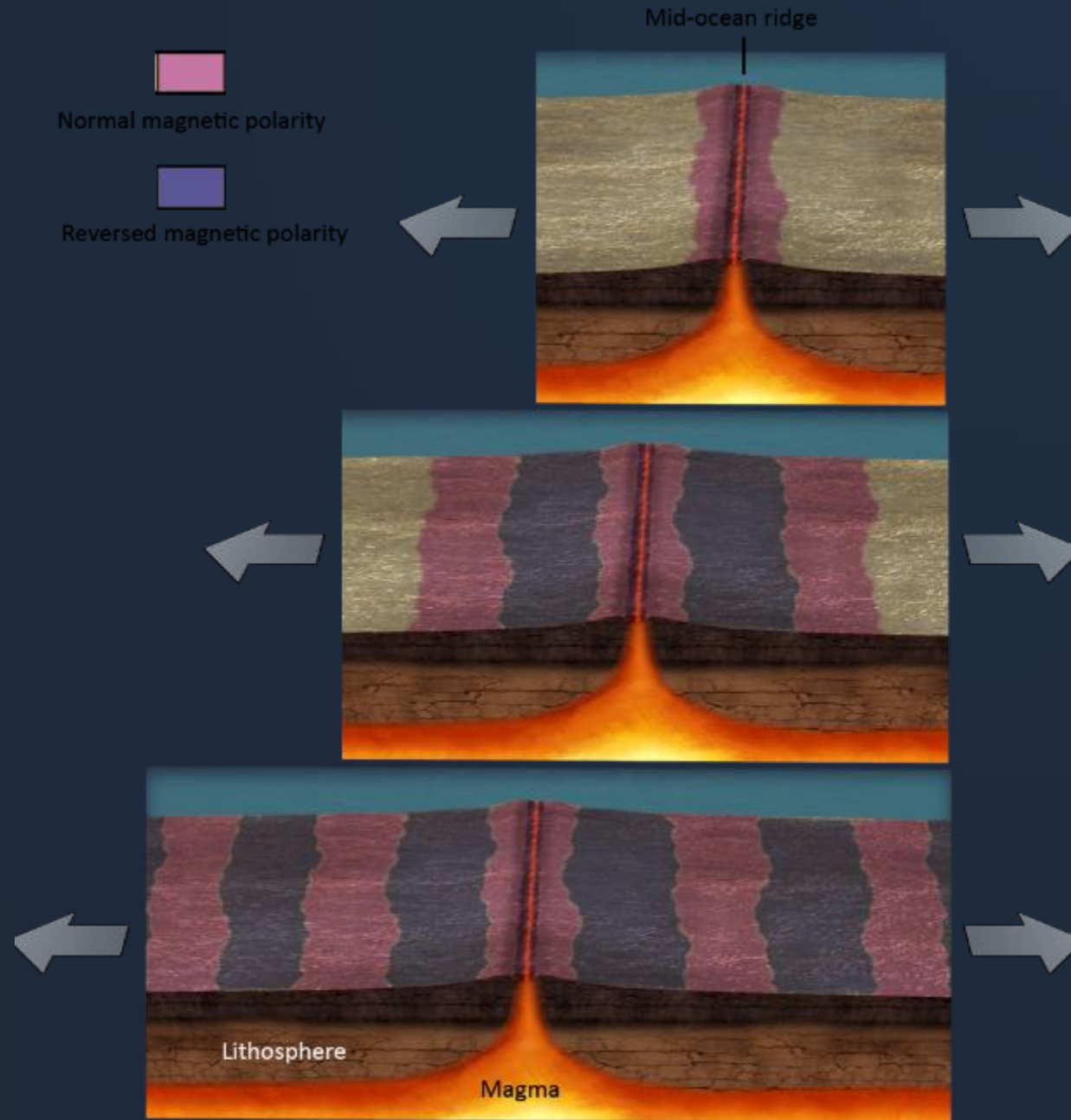
- Hotspots: The surface manifestation of a mantle plume is an area of volcanism, high heat flow, and crustal uplifting known as a hotspot.
- Hotspot Tracks: As a tectonic plate moves over a stationary mantle plume, a linear chain of volcanic structures forms.
- By measuring the age of the volcanoes in the chain (e.g., the Hawaiian Islands), geologists can precisely determine the direction and rate of plate motion over time.



Evidence: Paleomagnetism

Earth's magnetic field periodically reverses polarity. Iron-rich minerals in cooling basaltic lava align with the magnetic field, locking in a record of the polarity at the time of eruption.

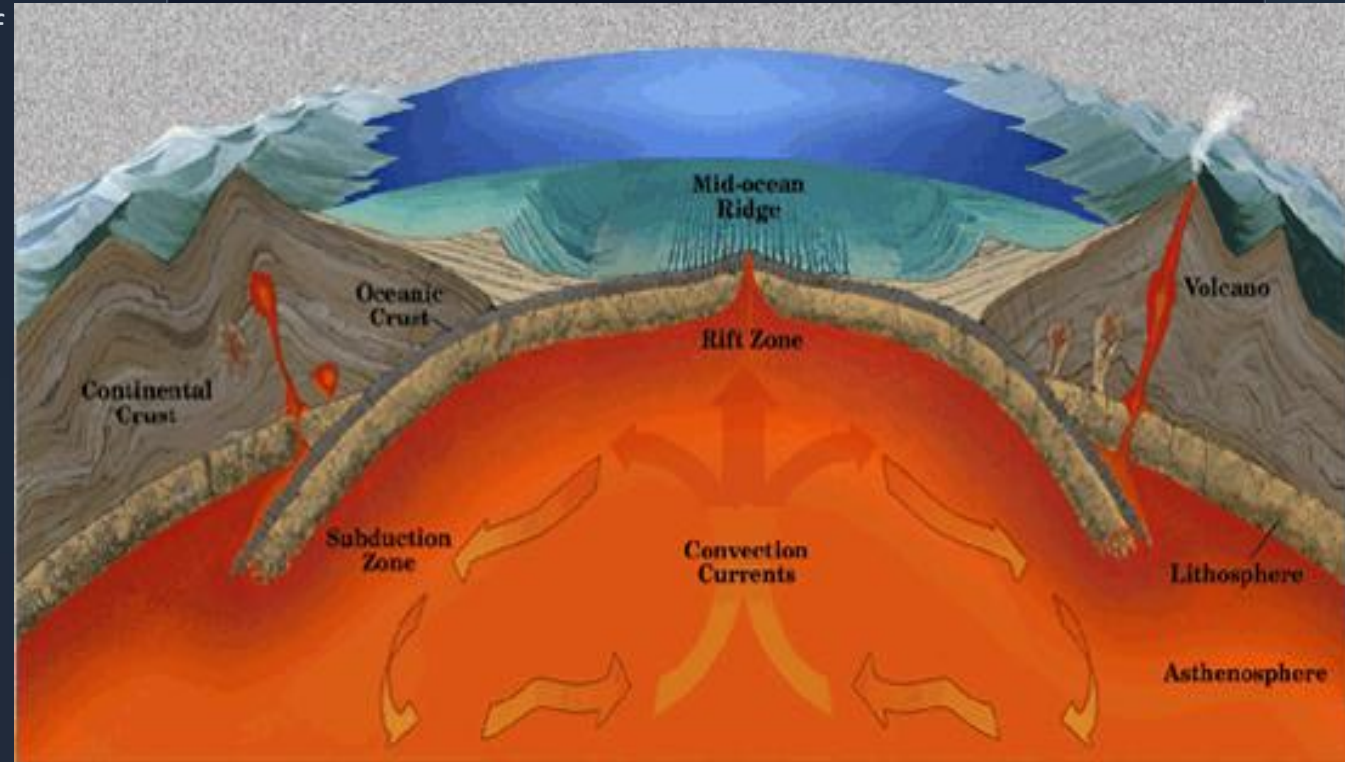
- **Magnetic Striping:** Magnetometers towed across the ocean floor revealed alternating stripes of high and low-intensity magnetism.
- **Symmetrical Patterns:** These stripes are perfectly symmetrical on either side of the mid-ocean ridges.
- This symmetry is definitive proof of seafloor spreading, showing identical crust formation and divergence over millions of years.



Mechanisms of Plate Motion

Convection in the mantle—where warm, buoyant rock rises and cooler, denser material sinks—is the fundamental driving force of plate tectonics.

- **The Convective Engine:** Earth's internal heat (from radioactive decay and residual planetary formation heat) drives this slow, creeping motion in the solid mantle.
- **Lithosphere Interaction:** The tectonic plates are the cool, rigid outermost layer of this convective system, driven laterally by complex forces applied at their boundaries and bases.



Driving Forces: Slab Pull & Ridge Push

Slab Pull

Considered the primary driver of plate motion. As oceanic lithosphere cools, it becomes denser than the underlying asthenosphere. When it subducts, gravity pulls the massive, cold slab downward, dragging the trailing plate along with it.

Ridge Push

A gravity-driven mechanism that operates at divergent boundaries. The elevated position of the mid-ocean ridge causes the newly formed, buoyant lithosphere to slide gravitationally down the flanks, gently pushing the plate ahead of it.

Models of Plate-Mantle Convection

Whole Mantle (Plume) Model

Proposes that cold oceanic lithosphere descends entirely to the core-mantle boundary. This downward flow stirs the entire 2,900-kilometer thickness of the mantle, balanced by rising plumes of hot rock originating from the core-mantle interface.

Layer Cake Model

Suggests that the mantle is divided into two distinct zones of convection. A dynamic upper layer (driven by subducting slabs) exists above a sluggish, isolated lower mantle. In this model, the layers do not mix significantly.

The Philippine Tectonics Setting

The Philippine archipelago is situated in one of the most structurally complex and seismically active regions on Earth.

- Tri-Plate Interaction: The region's tectonics are dominated by the ongoing collision and interaction of three major lithospheric plates:
 - The Eurasian Plate (Sunda Block)
 - The Pacific Plate (via the Philippine Sea Plate)
 - The Indo-Australian Plate
- This interaction forms a complex network of subduction zones, collision zones, and major wrench faults.

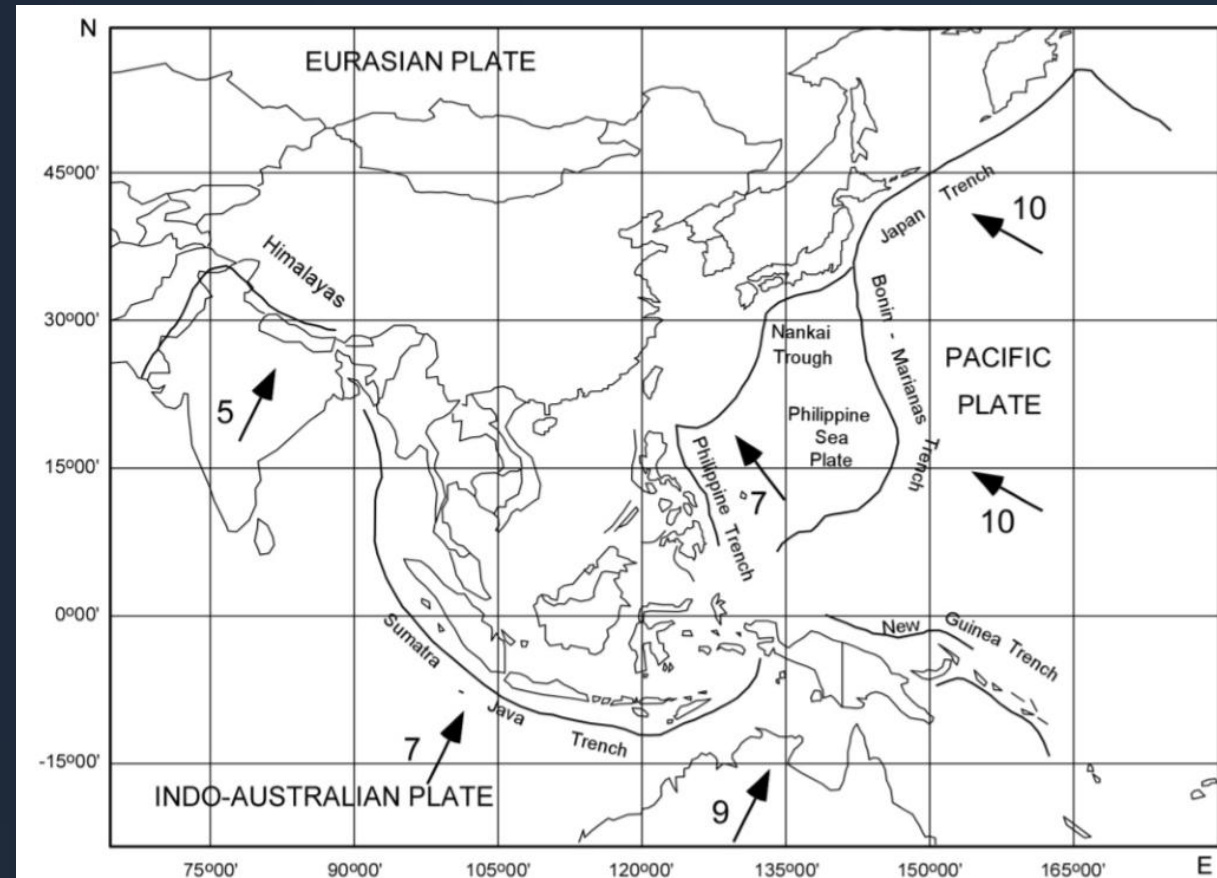


Fig. 1.1 Geodynamic setting of the Southeast Asia – West Pacific Domain. Numbers beside arrows indicate rates of plate motion in cm/yr relative to Eurasia. Modified from Barrier, 1985

Two Tectono-Stratigraphic Blocks

The Philippines can be geologically divided into two major terranes with distinct origins and seismic profiles.

1. Philippine Mobile Belt (PMB)

Defined by Gervasio (1967), the PMB is the seismically active, rapidly deforming portion of the archipelago. It comprises island arcs constrained by subduction zones of opposing polarities (e.g., the Philippine Trench to the east, Manila/Negros Trenches to the west).

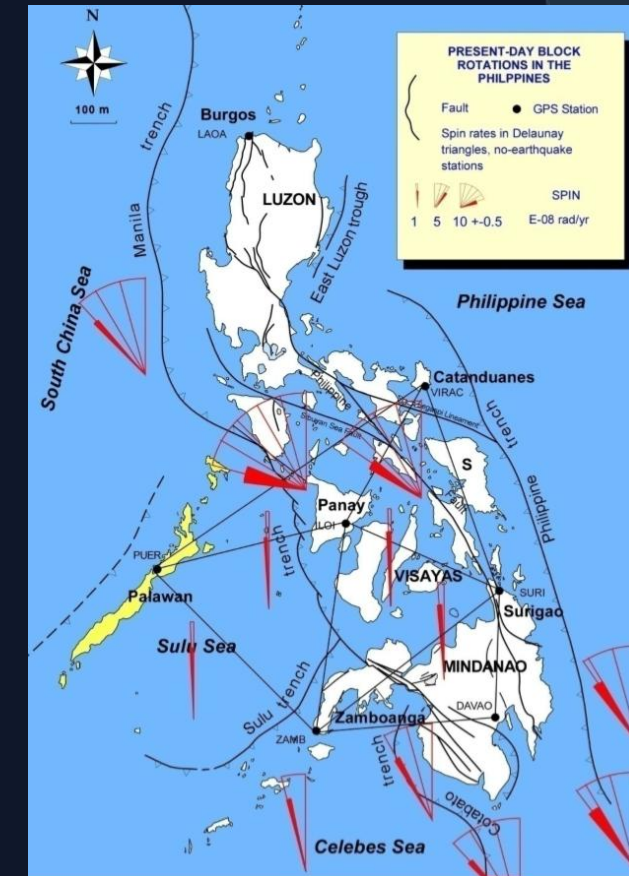
2. Palawan-Mindoro Continental Block (PCB)

An aseismic microcontinental block consisting of Palawan, Mindoro, Romblon, and the Busuanga Peninsula. Originating from the Eurasian plate (southeastern China), this block drifted southward and collided with the PMB, driving massive tectonic reorganization.

The Philippine Fault Zone

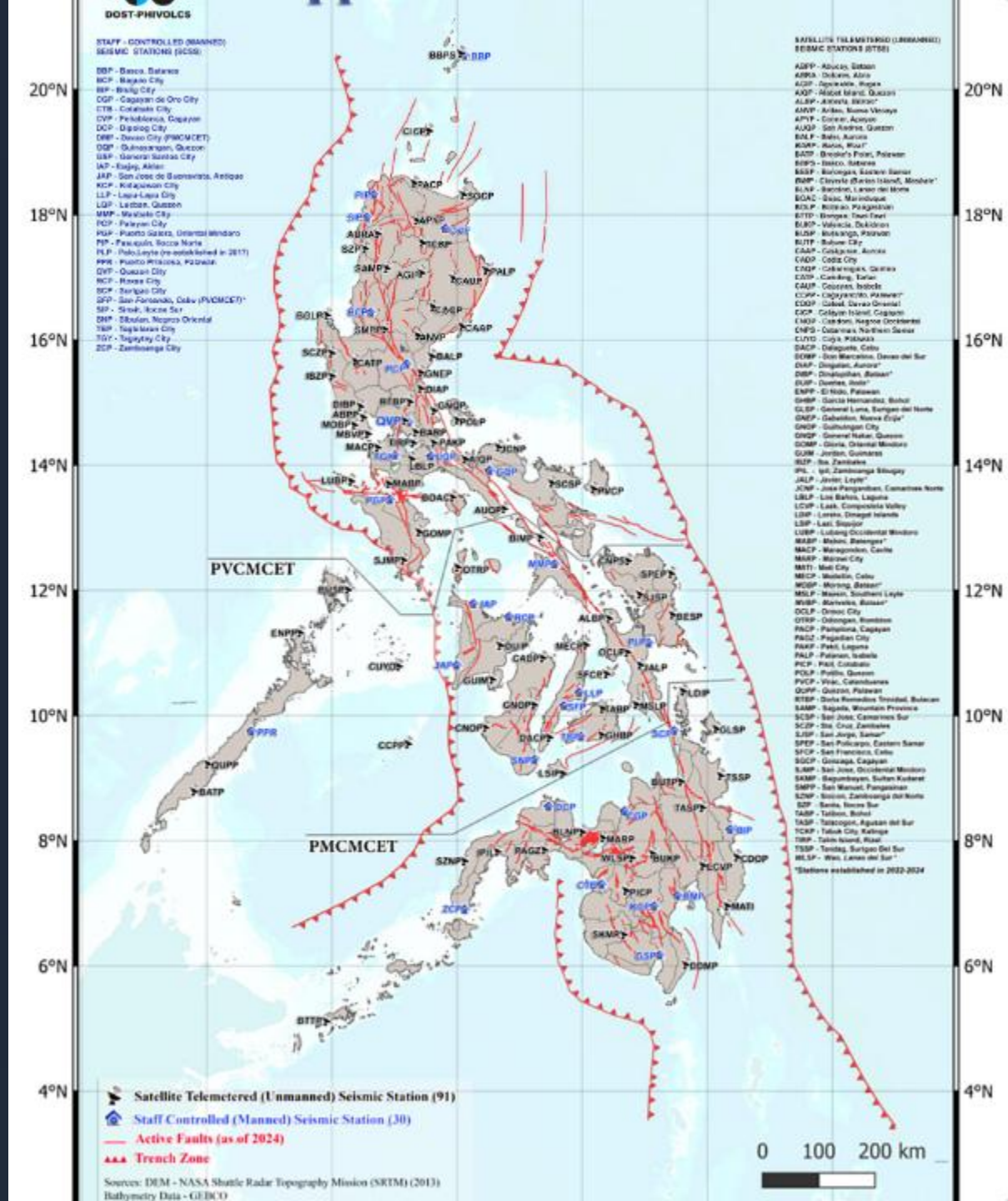
A major structural feature first detailed by Willis (1937), traversing the entire length of the Philippine Mobile Belt.

- Dimensions: Extends for more than 1,200 km from Luzon down to Eastern Mindanao.
- Kinematics: An active left-lateral strike-slip fault system in a shear-partitioning regime.
- Slip Rate: Moves at an average rate of 2 to 3 cm/year, accommodating the oblique convergence of the Philippine Sea Plate.
- Comparison: Its extensive geological significance is heavily compared to the San Andreas Fault in California.



- High Deformation Rates: GPS data indicates rapid block rotation and internal deformation within the PMB.

- **Tectonic Hazards:** This geologic setting is responsible for the region's devastating earthquakes, tsunamis, and prolific volcanism.



The Dynamic Earth

Plate Tectonics provides the unifying framework for understanding mountain building, volcanism, earthquakes, and the continuous evolution of the Earth's surface—with the Philippine archipelago standing as one of the most profound examples of this dynamic system in action.

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